SQL Database Integration — Assignment Documentation

This document compiles the execution results of the SQL steps performed in the pgAdmin environment for the FastAPI Calculator assignment.

Environment Overview

- **FastAPI Service**: Running on port `8000`
- **PostgreSQL Database**: Running on port `5432`
- **pgAdmin Management**: Accessible at `http://localhost:5050`

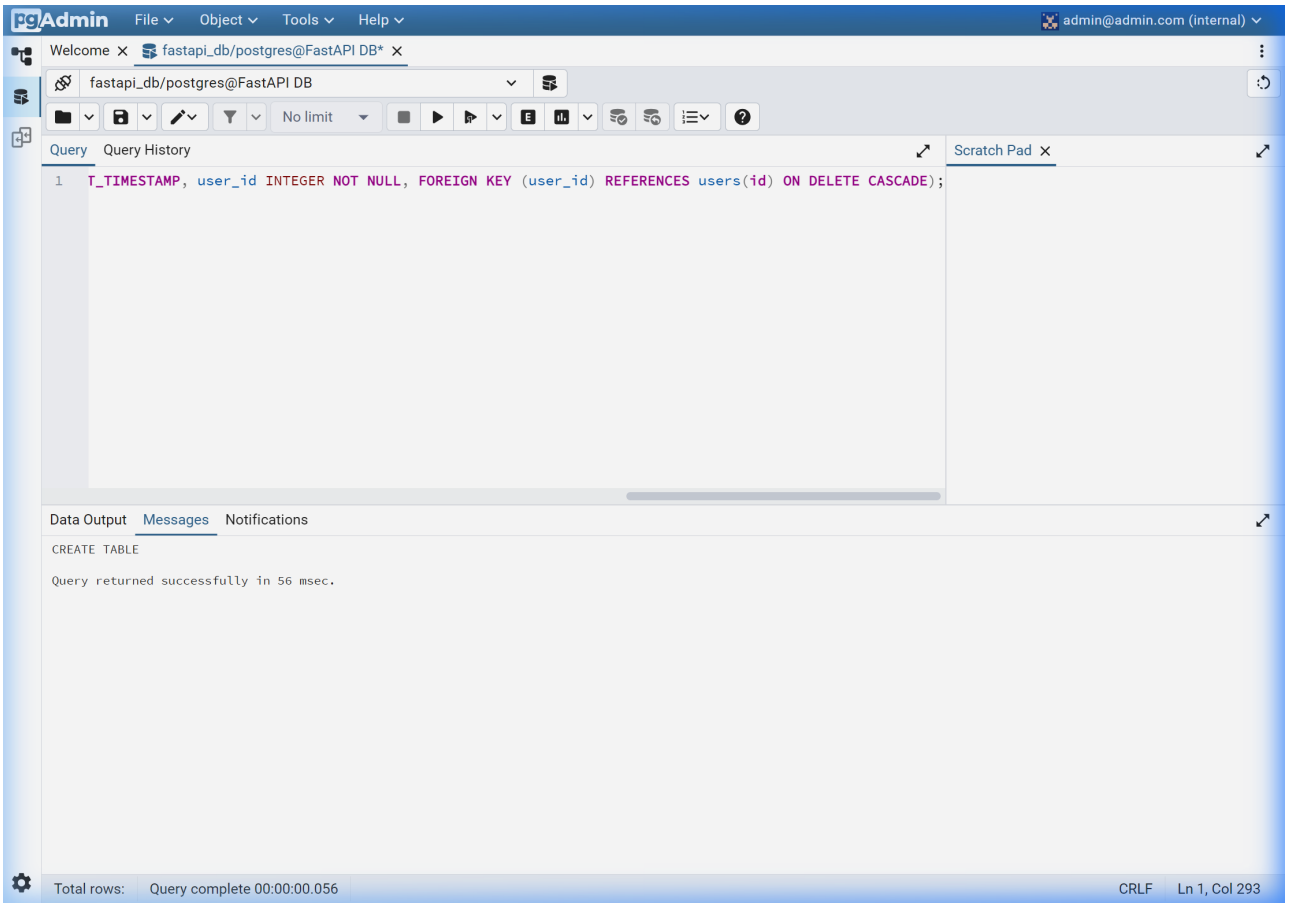
Repository Deliverables

The following files have been added to the repository to satisfy the assignment requirements:
- **`docker-compose.yml`**: Configures the FastAPI, PostgreSQL, and pgAdmin services.
- **`setup_db.sql`**: Contains all SQL commands for table creation, insertion, queries, updates, and deletions.
- **`pgadmin_servers.json`**: Enables auto-configuration of the `FastAPI DB` server in pgAdmin for immediate use.

SQL Step Execution Results

(A) Table Creation

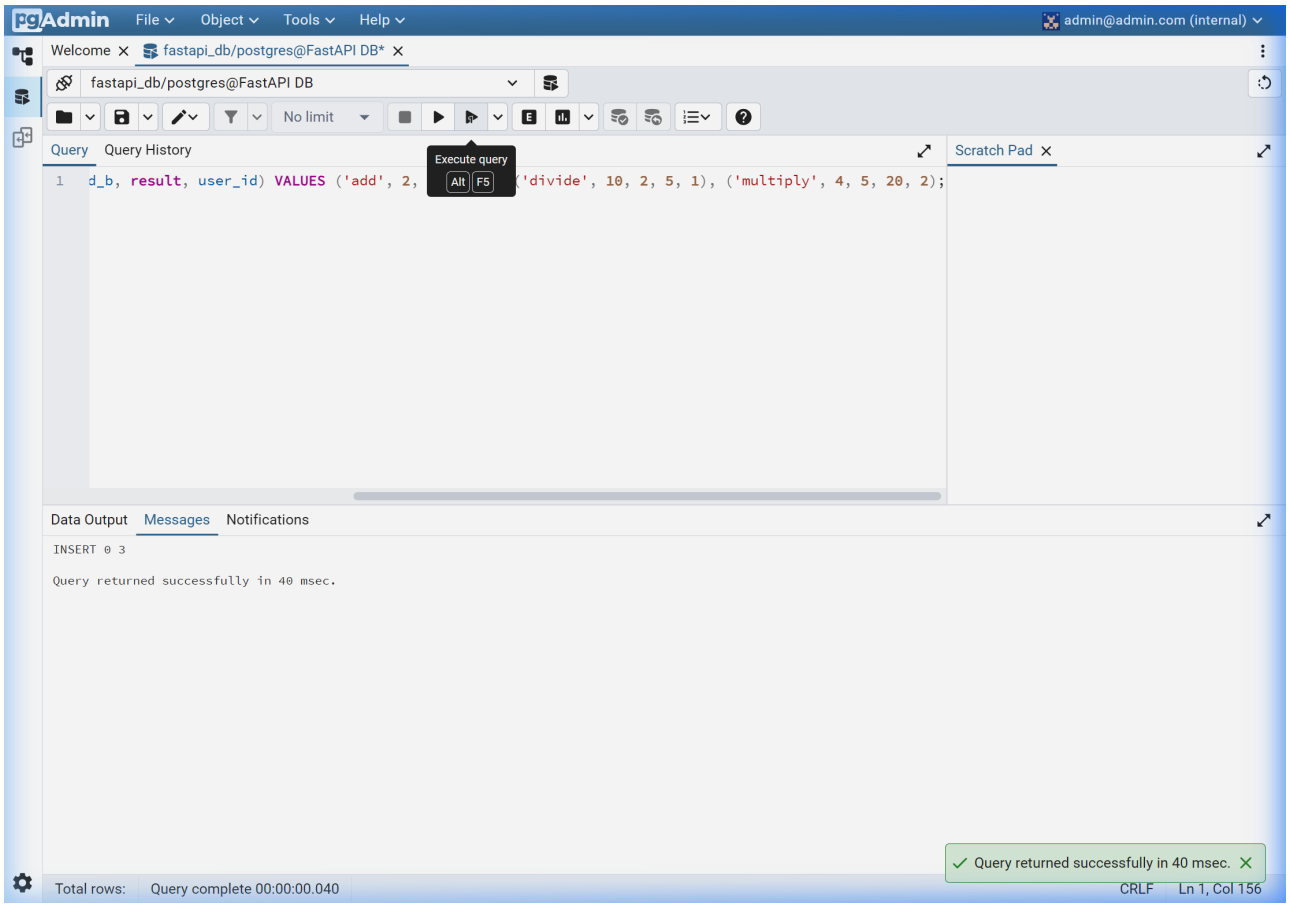
The `users` and `calculations` tables were successfully created with appropriate primary and foreign keys.



Caption: Tables created successfully using SERIAL primary keys and foreign key constraints.

(B) Data Insertion

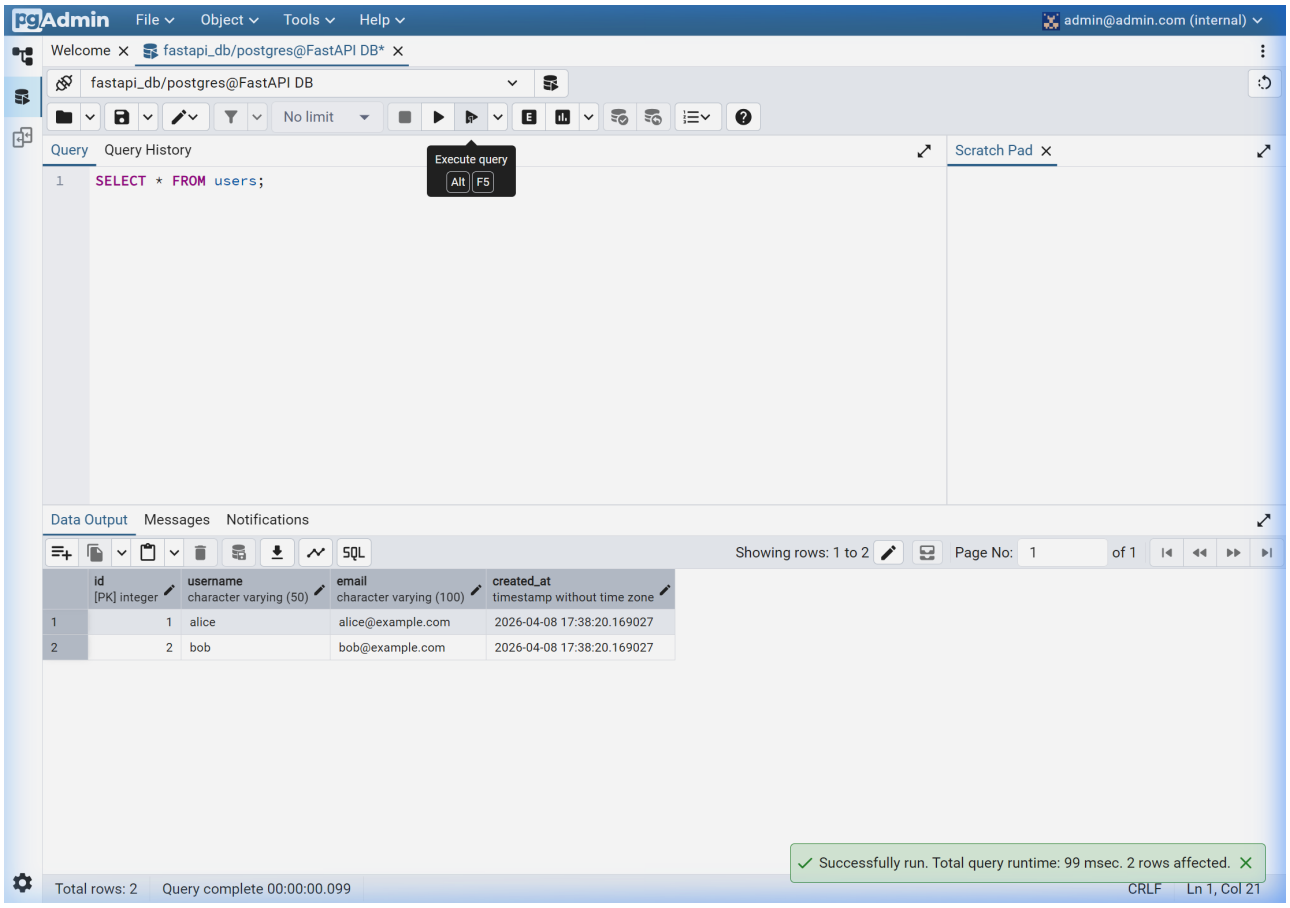
Two users ('alice', 'bob') and three calculation records were inserted.



Caption: Initial records inserted into both users and calculations tables.

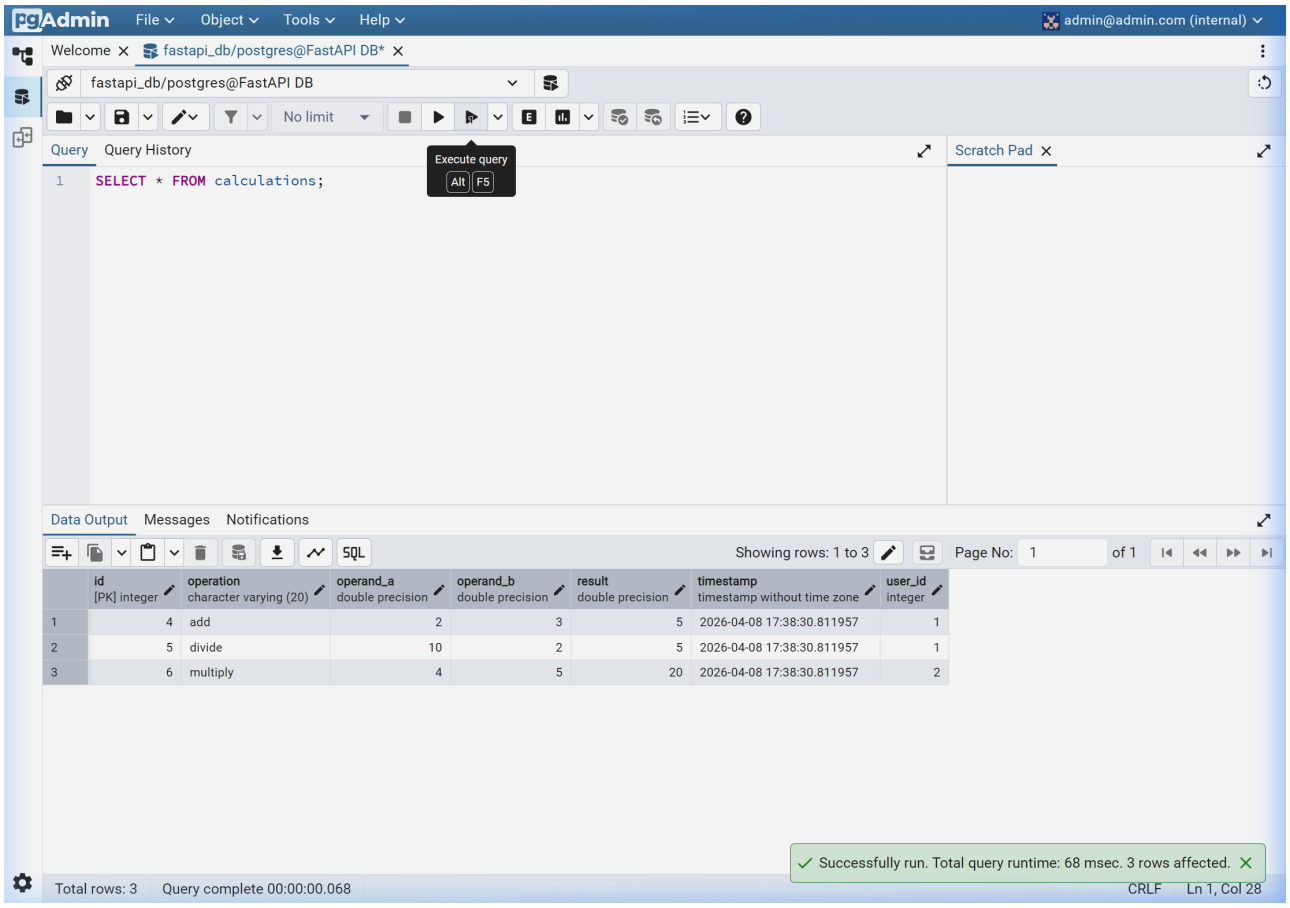
(C) Data Retrieval

All Users



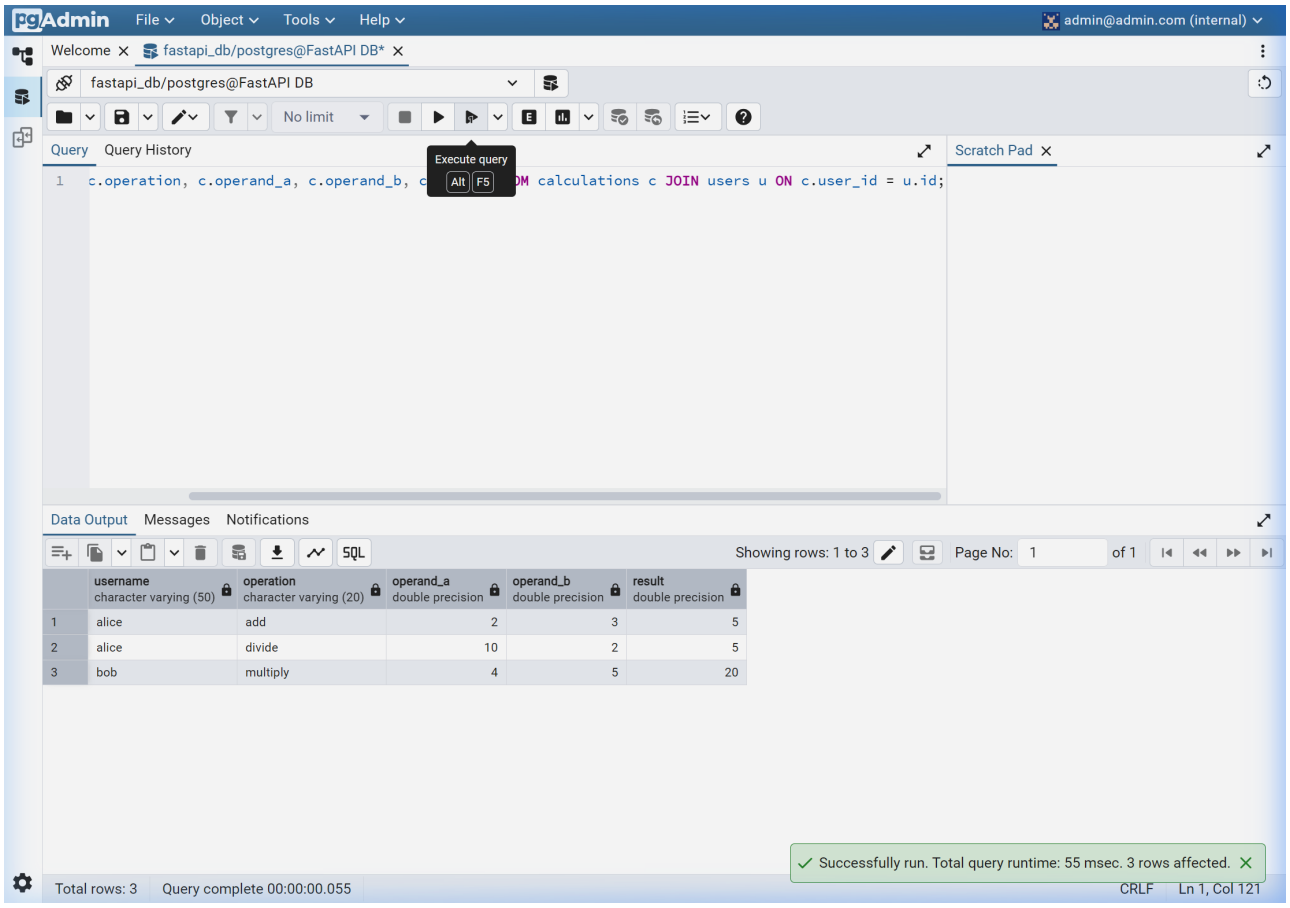
Caption: Retrieving all users from the users table.

All Calculations



Caption: Retrieving all calculations from the calculations table.

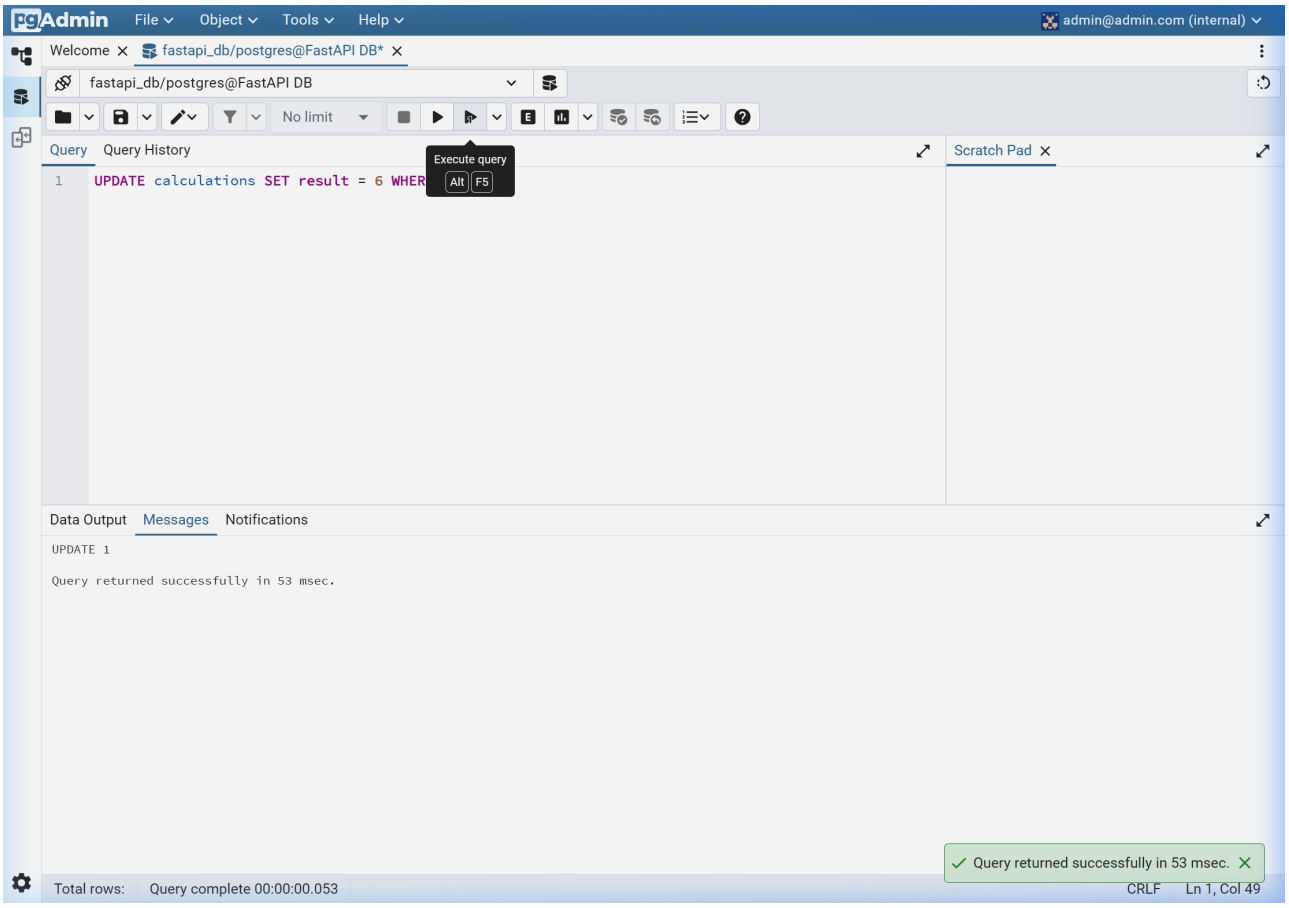
Joined Data



Caption: A JOIN operation showing the relationship between users and their specific calculations.

(D) Record Update

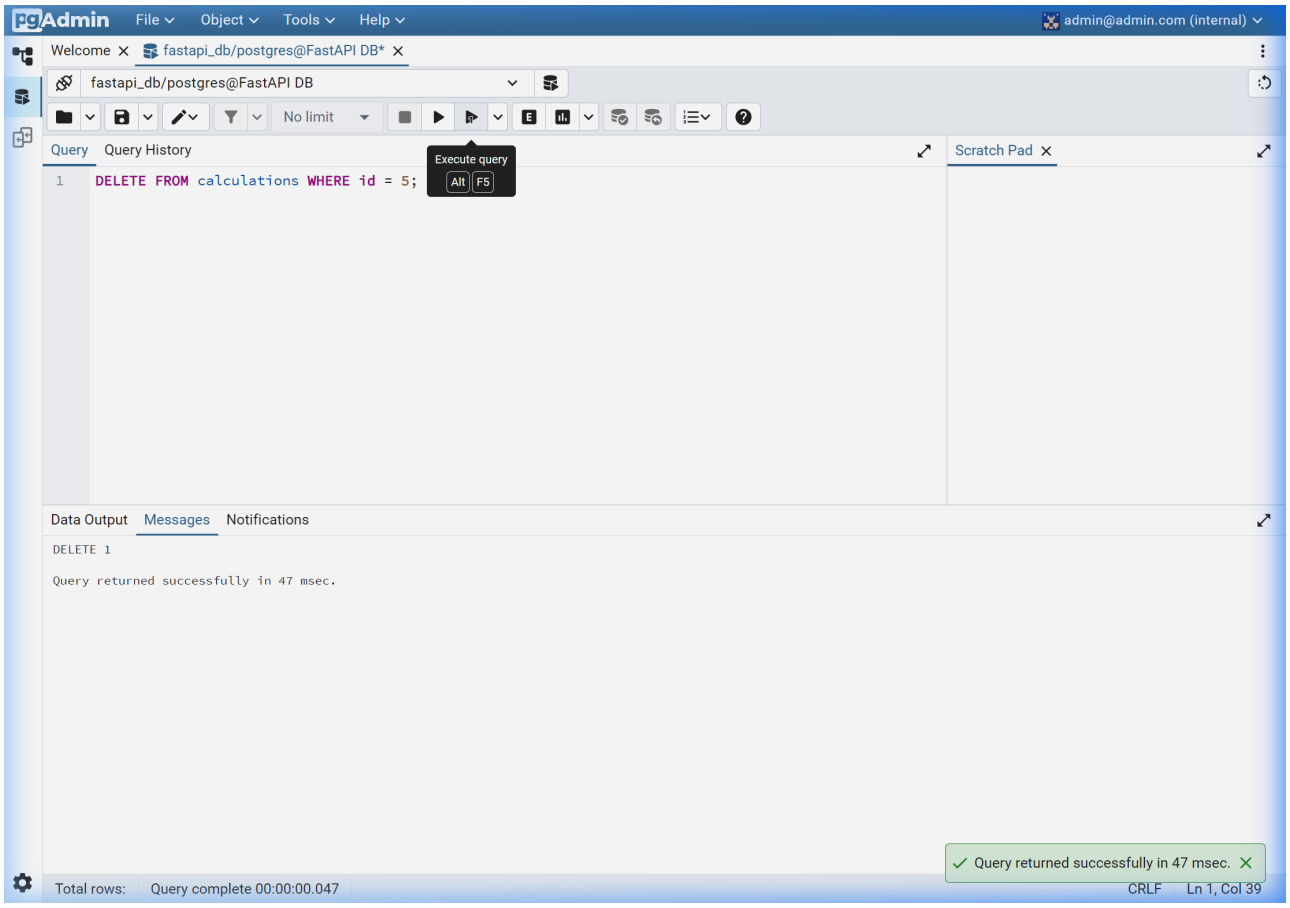
The result of a calculation record was updated to demonstration manipulation of existing data.



Caption: Successfully updated calculation result for record ID 4.

(E) Record Deletion

A calculation record was deleted to demonstrate data removal.



Caption: Successfully deleted record demonstrating the DELETE operation.

Conclusion

All SQL operations were completed successfully within the containerized environment. The use of Docker Compose ensured all services were correctly networked and accessible.